SL MINI PROJECT

Personal Wellbeing and Productivity Analyzer

ML-Based Productivity Prediction

1. Project Overview

This project develops a Personal Wellbeing and Productivity Analyzer that explores relationships between digital lifestyle habits, physical and mental wellbeing indicators, and productivity. The system performs exploratory data analysis, generates personalized habit recommendations based on comparison with dataset medians, and trains a supervised machine-learning model (Linear Regression) to predict a person's productivity score from lifestyle and wellbeing features.

Key objectives:

- Understand patterns linking sleep, stress, digital use, activity, and productivity.
- Provide individual-level reports and rule-based habit suggestions.
- Build and evaluate a simple regression model that estimates productivity_score from selected input features.

2. Dataset Description

Dataset name: digital_lifestyle_benchmark_2025.csv

The dataset contains 3,500 records and 24 columns describing demographic attributes, daily digital behaviour, physical activity, sleep, mental-health related scores, and productivity-related measures. There are no missing values and no duplicate rows after cleaning.

2.1 Column Summary

Column Group	Columns
Identifiers & Demographics	id, age, gender, region, income_level, education_level, daily_role
Digital Behaviour	device_hours_per_day, phone_unlocks, notifications_per_day, social_media_mins, device_type, digital_dependence_score
Lifestyle & Activity	study_mins, physical_activity_days, sleep_hours, sleep_quality
Wellbeing Scores	anxiety_score, depression_score, stress_level, happiness_score, focus_score, high_risk_flag
Target	productivity_score (continuous numerical score)

2.2 Key Descriptive Statistics

Selected average values across the full dataset:

Indicator	Mean (approx.)
Age	28.1 years
Device hours per day	7.32 hours
Phone unlocks	147
Social media (mins/day)	159 mins
Study minutes	108 mins
Physical activity days/week	3.35 days
Sleep hours	7.25 hours
Sleep quality (scale)	2.71
Stress level	5.08
Happiness score	6.42

Focus score	41.6
Productivity score	65.30 (std ≈ 9.66)
Digital dependence score	36.68

2.3 Categorical Distributions (high-level)

- Gender: Female ≈ 1,835; Male ≈ 1,665
- Regions: Europe, Asia, North America, Africa, South America, Middle East (Europe and Asia largest)
- Income: Low and Lower-Mid are the most frequent categories
- Education: Bachelor most common, followed by High School, Master, PhD
- Daily role: Full-time Employee (largest), Student, Part-time/Shift, Unemployed_Looking, Caregiver/Home
- Device type: relatively balanced across Android, Tablet, Laptop, iPhone

3. Steps Performed

3.1 Data Loading and Exploration

1. Imported required libraries (pandas, numpy, matplotlib, seaborn).
2. Loaded the CSV file and verified shape (3,500 × 24).
3. Inspected column names, data types, missing-value counts (all zero), and number of unique values.
4. Confirmed absence of duplicate rows.

3.2 Data Cleaning

5. Created a working copy (df_clean).
6. Removed any exact duplicate rows.
7. Filled potential missing numeric values with column medians (none were present in the source data).

3.3 Exploratory Analysis and Visualisation

8. Computed overall averages for key wellbeing and productivity indicators.
9. Generated histograms / distribution plots for sleep_hours, stress_level, happiness_score, focus_score, productivity_score and device_hours_per_day.
10. Calculated correlations between numeric features and productivity_score to identify associated indicators.
11. Compared average productivity across daily_role categories.
12. Summarised digital-habit statistics (means vs medians) for device hours, social media, unlocks and digital dependence.

3.4 Individual Report Generator

Implemented a function that accepts a person ID and prints a personalised summary of that individual's demographics, digital habits, sleep, activity, mental-health related scores, focus, productivity and digital-dependence values, together with comparison notes against dataset medians.

3.5 Rule-Based Habit Recommendation System

Developed a recommendation engine that compares a selected person's indicators with the corresponding dataset medians and emits practical suggestions when values fall below or above thresholds. Categories covered include:

- Sleep routine and sleep quality
- Focus techniques (e.g., Pomodoro-style sessions)
- Productivity prioritisation
- Study scheduling
- Stress and emotional wellbeing support
- Physical activity
- Screen time, social media, phone unlocks and digital balance

A clear disclaimer notes that suggestions are general, non-medical, and do not guarantee improved productivity.

3.6 Machine Learning – Productivity Prediction

A supervised regression pipeline was added to estimate productivity_score from lifestyle and wellbeing features.

Selected input features (X):

- sleep_hours, sleep_quality, physical_activity_days
- stress_level, happiness_score, focus_score
- study_mins, device_hours_per_day, digital_dependence_score

Target (y):

productivity_score

Process:

1. Prepared feature matrix and target vector; median-imputed any residual missing values.
2. Split data 80 % training / 20 % testing (random_state = 42).
3. Trained a LinearRegression model.
4. Evaluated on the held-out test set using Mean Absolute Error (MAE) and R^2 score.
5. Enabled single-person prediction by ID, reporting both predicted and actual productivity scores.

4. Analysis and Key Findings

4.1 Overall Patterns

- Average productivity score is approximately 65.3 with a standard deviation of ~9.7, indicating moderate variation across individuals.
- Typical sleep is close to 7.25 hours; average stress sits near the middle of the observed range.
- Digital usage is substantial: mean device time ≈ 7.3 h/day, social-media time ≈ 2.7 h/day, and frequent phone unlocks.
- Physical activity averages roughly 3.3 days per week, leaving room for improvement for many individuals.

4.2 Role-based Productivity

Average productivity differs across daily roles. The analysis ranked roles by mean productivity_score, highlighting which occupational / lifestyle groups tend to report higher or lower scores. These group averages describe central tendencies only and do not apply to every individual.

4.3 Digital Habits

For several digital indicators the mean exceeds the median, suggesting a right-skewed distribution (a minority of users with very high usage pull the average upward). This pattern is common for screen-time and notification metrics.

4.4 Machine-Learning Model Performance

On the 20 % test set the Linear Regression model achieved:

- Mean Absolute Error (MAE) ≈ 7.07
- R^2 score ≈ 0.00

An R^2 near zero indicates that the linear combination of the chosen features explains little of the variance in productivity_score on unseen data. The absolute error of roughly 7 points on a scale whose standard deviation is ~9.7 shows the model's predictions are only modestly better than guessing the mean. Possible reasons include weak linear relationships, missing important predictors, non-linear effects, or noise in the target variable. The model remains useful as an educational demonstration of a full supervised-learning pipeline (feature selection, train/test split, training, evaluation, and single-record inference).

4.5 Important Limitations

- Correlations and group averages do not imply causation.
- Recommendations are general lifestyle suggestions, not medical or psychological advice.
- The linear model is intentionally simple; more advanced algorithms or richer feature engineering would be needed for stronger predictive performance.
- Results are specific to this synthetic / benchmark dataset and should not be generalised without further validation.

5. Technologies and Tools Used

- Python 3
- pandas & numpy – data loading, cleaning and aggregation
- matplotlib & seaborn – exploratory visualisations
- scikit-learn – train_test_split, LinearRegression, mean_absolute_error, r2_score
- Jupyter / Google Colab notebook environment

6. Conclusion

The Personal Wellbeing and Productivity Analyzer successfully integrates descriptive analytics, individual reporting, rule-based habit recommendations and a basic machine-learning prediction component. The project demonstrates an end-to-end workflow—from data exploration through supervised regression—while emphasising responsible interpretation: findings describe patterns present in the dataset and must not be treated as diagnostic or causal claims. Future extensions could explore non-linear models, additional features, interaction terms, or clustering of lifestyle profiles to improve both insight generation and predictive accuracy.